

ARID DESERT LANDFORMS / THAR DESERTIFICATION

CONCEPT FIREWALL → DESERT-FORMING CONTROLS → AEOLIAN TRANSPORT → DUNE DISCRIMINATOR → WATER IN THE DESERT → THAR SPATIAL ANATOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

CONCEPT FIREWALL

CONCEPT FIREWALLCLIMATIC REGIONLONG-TERM MOISTURE DEFICITTEMPORARY ANOMALYDEGRADATION WITHIN DRYLANDSMUST REMEMBERARID GEOMORPHOLOGY LINKS MOISTURE DEFICIT, SPARSE COVER, MECHANICAL WEATHERING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
DESERT → climatic region	DROUGHT → temporary anomaly	MUST REMEMBER: Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration
ARIDITY → long-term moisture deficit	DESERTIFICATION → degradation within drylands	pediment–bajada–playa and Thar land-use pressures frame desertification as land degradation.

CORE CLAIM

- DESERT → climatic region
- ARIDITY → long-term moisture deficit

MECHANISM

- DROUGHT → temporary anomaly
- DESERTIFICATION → degradation within drylands

CONSEQUENCE / LIMIT

- MUST REMEMBER: Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration
- pediment–bajada–playa and Thar land-use pressures frame desertification as land degradation.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration; pediment–bajada–playa and Thar land-use pressures frame desertification as land degradation.

01

STAGE 01

DESERT-FORMING CONTROLS

DESERT-FORMING CONTROLS

SUBTROPICAL HIGHDESCENDING DRY AIRRAIN SHADOWMOISTURE LOSSCOLD CURRENTSTABLE DRY COASTAL AIRPOLAR COLD

SUBTROPICAL HIGH → descending dry airCONTINENTALITY / RAIN SHADOW → moisture lossCOLD CURRENT → stable dry coastal airPOLAR COLD → low absolute moisture

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SUBTROPICAL HIGH → descending dry air	COLD CURRENT → stable dry coastal air	SUBTROPICAL HIGH → descending dry air
CONTINENTALITY / RAIN SHADOW → moisture loss	POLAR COLD → low absolute moisture	CONTINENTALITY / RAIN SHADOW → moisture loss

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COLD CURRENT → stable dry coastal air; POLAR COLD → low absolute moisture.

02

STAGE 02

AEOLIAN TRANSPORT

AEOLIAN TRANSPORT

COARSE GRAINS ROLL OR SLIDESAND GRAINS BOUNCEFINE DUST TRAVELS FARVELOCITY OR CARRYING POWER FALLS

CREEP → coarse grains roll or slideSALTATION → sand grains bounceSUSPENSION → fine dust travels farDEPOSITION → velocity or carrying power falls

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CREEP → coarse grains roll or slide	SUSPENSION → fine dust travels far	CREEP → coarse grains roll or slide
SALTATION → sand grains bounce	DEPOSITION → velocity or carrying power falls	SALTATION → sand grains bounce

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUSPENSION → fine dust travels far; DEPOSITION → velocity or carrying power falls.

03

STAGE 03

DUNE DISCRIMINATOR

DUNE DISCRIMINATOR

LIMITED SAND, HORNS DOWNWINDABUNDANT SAND, ONE DOMINANT WINDALIGNED WITH RESULTANT WINDSVEGETATION ANCHORMULTIDIRECTIONAL WINDS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
BARCHAN → limited sand, horns downwind	LONGITUDINAL → aligned with resultant winds	BARCHAN → limited sand, horns downwind
TRANSVERSE → abundant sand, one dominant wind	PARABOLIC / STAR → vegetation anchor / multidirectional winds	TRANSVERSE → abundant sand, one dominant wind

CORE CLAIM

- BARCHAN → limited sand, horns downwind
- TRANSVERSE → abundant sand, one dominant wind

MECHANISM

- LONGITUDINAL → aligned with resultant winds
- PARABOLIC / STAR → vegetation anchor / multidirectional winds

CONSEQUENCE / LIMIT

- BARCHAN → limited sand, horns downwind
- TRANSVERSE → abundant sand, one dominant wind

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03

SALTATION → sand grains bounce

DEPOSITION → velocity or carrying power falls

SALTATION → sand grains bounce

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUSPENSION → fine dust travels far; DEPOSITION → velocity or carrying power falls.

03

STAGE 03

DUNE DISCRIMINATOR

DUNE DISCRIMINATOR

LIMITED SAND, HORNS DOWNWIND

ABUNDANT SAND, ONE DOMINANT WIND

ALIGNED WITH RESULTANT WINDS

VEGETATION ANCHOR

MULTIDIRECTIONAL WINDS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
BARCHAN → limited sand, horns downwind	LONGITUDINAL → aligned with resultant winds	BARCHAN → limited sand, horns downwind
TRANSVERSE → abundant sand, one dominant wind	PARABOLIC / STAR → vegetation anchor / multidirectional winds	TRANSVERSE → abundant sand, one dominant wind

CORE CLAIM

• BARCHAN → limited sand, horns downwind

• TRANSVERSE → abundant sand, one dominant wind

MECHANISM

• LONGITUDINAL → aligned with resultant winds

• PARABOLIC / STAR → vegetation anchor / multidirectional winds

CONSEQUENCE / LIMIT

• BARCHAN → limited sand, horns downwind

• TRANSVERSE → abundant sand, one dominant wind

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PARABOLIC / STAR → vegetation anchor / multidirectional winds.

04

STAGE 04

WATER IN THE DESERT

WATER IN THE DESERT

STORM RUNOFF

WADI DISCHARGE

MOUNTAIN FRONT

ALLUVIAL FAN

FANS COALESCE

CLOSED BASIN

PLAYA AND SALT CONCENTRATION

STORM RUNOFF → wadi discharge

MOUNTAIN FRONT → alluvial fan

FANS COALESCE → bajada

CLOSED BASIN → playa and salt concentration

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• STORM RUNOFF → wadi discharge	• FANS COALESCE → bajada	• STORM RUNOFF → wadi discharge
• MOUNTAIN FRONT → alluvial fan	• CLOSED BASIN → playa and salt concentration	• MOUNTAIN FRONT → alluvial fan

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FANS COALESCE → bajada; CLOSED BASIN → playa and salt concentration.

05

STAGE 05

THAR SPATIAL ANATOMY

THAR SPATIAL ANATOMY

SANDY MARUSTHALI CORE

BAGAR AND ARAVALLI TRANSITION

LUNI AND SALINE-BASIN ROUTE

CANAL-ALLUVIAL TRANSFORMATION

EAST → Bagar and Aravalli transition

WEST → sandy Marusthali core

SOUTHWEST → Luni and saline-basin route

NORTH → canal-alluvial transformation

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• WEST → sandy Marusthali core	• SOUTHWEST → Luni and saline-basin route	• WEST → sandy Marusthali core
• EAST → Bagar and Aravalli transition	• NORTH → canal-alluvial transformation	• EAST → Bagar and Aravalli transition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOUTHWEST → Luni and saline-basin route; NORTH → canal-alluvial transformation.

06

STAGE 06

SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND

NATURAL STRESS

COVER LOSS + SOIL DISTURBANCE

WATER MISMANAGEMENT

SALINITY OR DEPLETION

THRESHOLD CROSSED

RECOVERY CAPACITY DECLINES

VARIABLE RAIN + WIND → natural stress

COVER LOSS + SOIL DISTURBANCE → exposure

WATER MISMANAGEMENT → salinity or depletion

THRESHOLD CROSSED → recovery capacity declines

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• VARIABLE RAIN + WIND → natural stress	• WATER MISMANAGEMENT → salinity or depletion	• VARIABLE RAIN + WIND → natural stress
• COVER LOSS + SOIL DISTURBANCE → exposure	• THRESHOLD CROSSED → recovery capacity declines	• COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

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06

STAGE 06 SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND NATURAL STRESS COVER LOSS + SOIL DISTURBANCE WATER MISMANAGEMENT SALINITY OR DEPLETION THRESHOLD CROSSED RECOVERY CAPACITY DECLINES



CORE CLAIM

- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

MECHANISM

- WATER MISMANAGEMENT → salinity or depletion
- THRESHOLD CROSSED → recovery capacity declines

CONSEQUENCE / LIMIT

- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

07

STAGE 07 ARAVALLI BOUNDARY

ARAVALLI BOUNDARY MODIFIES LOCAL WIND AND RUNOFF DIVIDES AND ROUTES WATER SUPPORTS CONNECTIVITY NOT AN IMPERMEABLE DESERT WALL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
RELIEF → modifies local wind and runoff	HABITAT → supports connectivity	RELIEF → modifies local wind and runoff
DRAINAGE → divides and routes water	LIMIT → not an impermeable desert wall	DRAINAGE → divides and routes water

CORE CLAIM

- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

MECHANISM

- HABITAT → supports connectivity
- LIMIT → not an impermeable desert wall

CONSEQUENCE / LIMIT

- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HABITAT → supports connectivity; LIMIT → not an impermeable desert wall.

08

STAGE 08 CANAL BALANCE

CANAL BALANCE WATER SUPPLY AND PRODUCTION WATER-TABLE RISE POOR DRAINAGE WATERLOGGING AND SALINITY CROP-WATER FIT + DRAINAGE + MONITORING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
BENEFIT → water supply and production	POOR DRAINAGE → waterlogging and salinity	BENEFIT → water supply and production
SEEPAGE → water-table rise	MANAGEMENT → crop-water fit + drainage + monitoring	SEEPAGE → water-table rise

CORE CLAIM

- BENEFIT → water supply and production
- SEEPAGE → water-table rise

MECHANISM

- POOR DRAINAGE → waterlogging and salinity
- MANAGEMENT → crop-water fit + drainage + monitoring

CONSEQUENCE / LIMIT

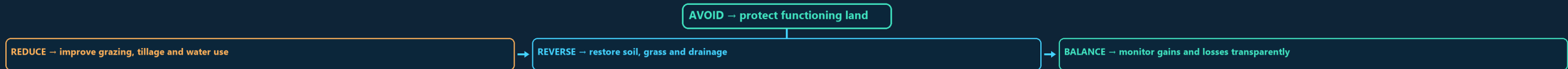
- BENEFIT → water supply and production
- SEEPAGE → water-table rise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGEMENT → crop-water fit + drainage + monitoring.

09

STAGE 09 LDN RESPONSE HIERARCHY

LDN RESPONSE HIERARCHY PROTECT FUNCTIONING LAND IMPROVE GRAZING, TILLAGE AND WATER USE RESTORE SOIL, GRASS AND DRAINAGE MONITOR GAINS AND LOSSES TRANSPARENTLY CLOSE DISTINCTION DESERT IS A CLIMATIC MOISTURE CONDITION, DESERTIFICATION IS DEGRADATION



CORE CLAIM

- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

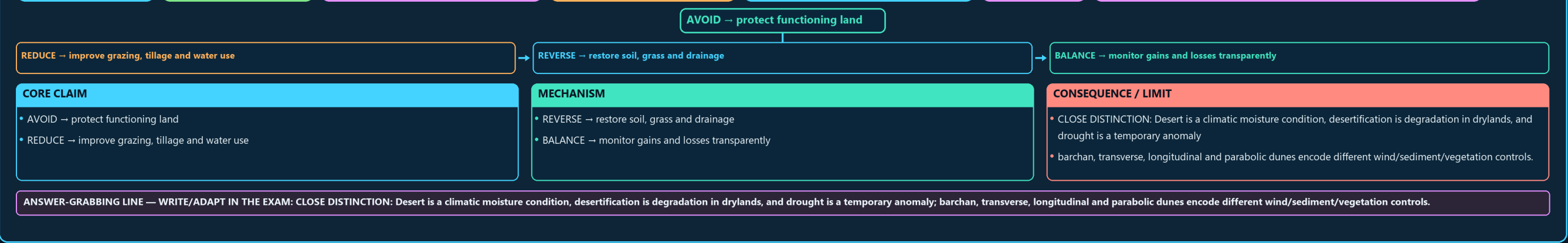
MECHANISM

- REVERSE → restore soil, grass and drainage
- BALANCE → monitor gains and losses transparently

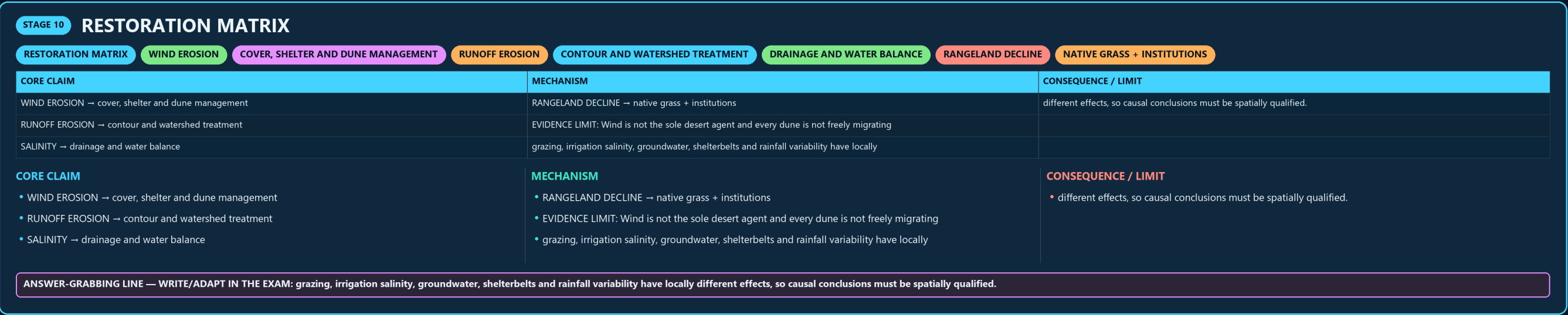
CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly
- barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.

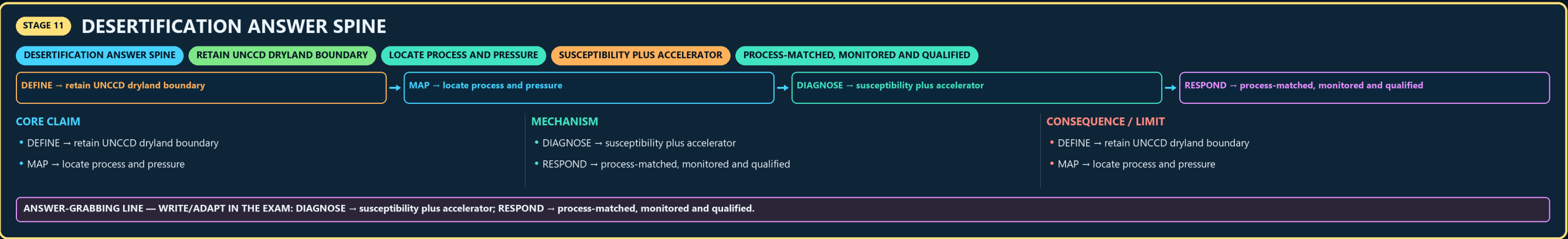
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.



10



11



E

